

Task 1: Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



The diagram above shows the 8 pins of an RJ-45 connector (pin 1 on the left, clip facing away, gold contacts up) with the wire color assigned to each pin under both wiring standards.

Pin-by-pin colors:

- T568A — Pin 1: White/Green, Pin 2: Green, Pin 3: White/Orange, Pin 4: Blue, Pin 5: White/Blue, Pin 6: Orange, Pin 7: White/Brown, Pin 8: Brown
- T568B — Pin 1: White/Orange, Pin 2: Orange, Pin 3: White/Green, Pin 4: Blue, Pin 5: White/Blue, Pin 6: Green, Pin 7: White/Brown, Pin 8: Brown

The only difference between the two standards is that the orange and green pairs are swapped; the blue pair (pins 4 and 5) stays the same in both.

Task 2: Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Standard	Speed	Cable type	Max cable length	Connector
10BASE-T	10 Mbps	Cat3 or better (UTP)	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better (UTP)	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or better (UTP)	100 m	RJ-45

Note: 10BASE-T and 100BASE-TX only use 2 of the 4 wire pairs (pins 1, 2, 3, 6), while 1000BASE-T uses all 4 pairs simultaneously — this is how it achieves gigabit speeds over the same cable length and connector.

Task 3: Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

A crossover cable is used — one end wired to the T568A standard and the other end wired to T568B.

Reason: Think of an Ethernet cable like a pair of walkie-talkies where one wire is only for **talking (transmitting)** and the other wire is only for **listening (receiving)**.

- **The Problem with a Normal (Straight-Through) Cable:** A standard cable connects the "talking" pin on Laptop A directly to the "talking" pin on Laptop B. If both laptops try to talk into the same wire, neither one is listening to it. They just blast data at each other on the same line, and no communication happens.
- **The Solution (Crossover Cable):** A crossover cable is physically wired to cross the lines in the middle. It takes Laptop A's "talking" wire and plugs it directly into Laptop B's "listening" wire.

Usually, a network switch or internet router handles this traffic-cop crossing logic for you internally. When you remove the switch and plug two laptops directly into each other, the cable has to do the crossing manually.

Note: Most modern network interface cards (post-2000s) support **Auto-MDI-X**, which automatically detects and electronically swaps transmit/receive pairs as needed. Because of this, a straight-through cable often works for a direct connection today as well. However, the crossover cable remains the technically correct answer, since it explains the underlying wiring principle being tested.

Task 4: List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Likely Ethernet standard	Reason
Home Wi-Fi router (Jio Fiber ONT/router)	1000BASE-T	High-performance devices require massive bandwidth for downloading massive modern games, low-latency online gaming, and local network file sharing
Smart TV or Streaming Box	100BASE-TX	most mainstream television manufacturers (such as Samsung and LG) still limit their physical Ethernet ports to 100 Mbps to save on production costs. Because streaming a high-quality 4K HDR movie only requires about 25 to 50 Mbps of bandwidth, the 100 Mbps maximum of a 100BASE-TX connection is plenty for video playback
HP DJ 2130 printer	100BASE-TX	Printers and older machines need modest throughput, so cheaper Fast Ethernet chips are common